

Hands-on project Report on

Title: Automatic headlight control using LDR(ADC)

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requirements for the course Technical Training
(22TP610)

Executed by Team members

NAME	USN
Maithrayi S.V	1NT23EC077
Rakshita S.W	1NT23EC124
Samvartha G P	1NT23EC140
Nandan K	1NT23EC090
Samanvith	1NT23CS078

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1. Problem statement

In conventional vehicles, the operation of headlights is manually controlled by the driver. This manual system often leads to several practical problems. During night time, foggy weather, rainy conditions, or while entering tunnels, drivers may forget to switch on the headlights, resulting in poor visibility and increasing the risk of accidents. Conversely, drivers sometimes forget to turn off the headlights during daytime, causing unnecessary battery drainage and reducing the life of the headlight bulbs.

These issues highlight the need for an intelligent system that can automatically detect ambient light conditions and control the headlights without human intervention. The existing manual headlight control is unreliable, especially for new or inexperienced drivers, and in situations where the driver is distracted or fatigued.

To address these challenges, this project presents an **Automatic Headlight Control System using LDR and ADC**. The system continuously monitors the intensity of surrounding light using a Light Dependent Resistor (LDR) sensor. The analog output of the LDR is converted into digital form by the Analog-to-Digital Converter (ADC) present in the microcontroller. Based on the processed data, the system automatically switches the headlights **ON** when it is dark and **OFF** when sufficient daylight is available.

This prototype, as demonstrated in the video, successfully shows the working of the system by turning the headlight LED **ON** when the LDR is covered (simulating darkness) and **OFF** when exposed to light.

2. Objectives

- ❖ To design and develop an automatic headlight control system that eliminates the need for manual operation of vehicle lights.
- ❖ To detect ambient light intensity using a Light Dependent Resistor (LDR) sensor.
- ❖ To convert the analog signal from LDR into digital form using Analog-to-Digital Converter (ADC) for processing.
- ❖ To automatically switch **ON** the headlights (LED) when the surrounding light intensity falls below a set threshold (dark condition) and switch them **OFF** during daylight.
- ❖ To demonstrate the real-time working of the system using a microcontroller-based prototype.
- ❖ To improve road safety by reducing human errors and ensure efficient use of vehicle battery by preventing unnecessary headlight usage.

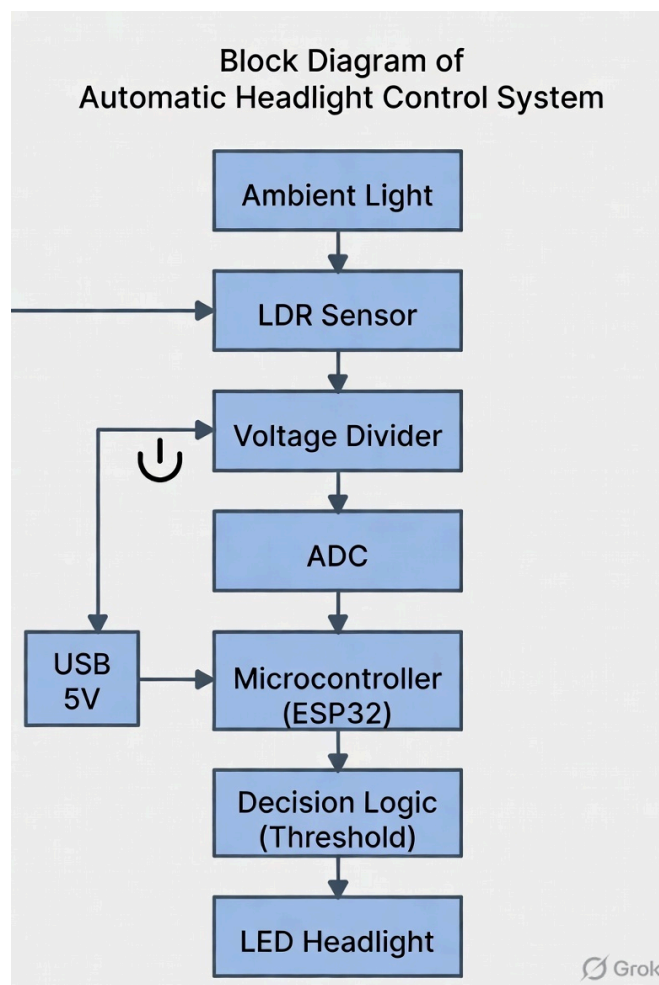
3. Design

The Automatic Headlight Control System is designed as a simple, low-cost microcontroller-based prototype. The system mainly consists of an LDR (Light Dependent Resistor) as the light sensing element, a microcontroller with built-in ADC, and an LED acting as the headlight.

3.1 System Design:

The design is based on a closed-loop control system where ambient light is continuously sensed and processed to control the output (headlight).

3.2 Block Diagram:



- **Input:** LDR Sensor (detects light intensity)
- **Signal Processing:** Voltage Divider → ADC (Analog to Digital Converter) → Microcontroller
- **Decision Making:** Microcontroller compares ADC value with predefined threshold
- **Output:** LED (Headlight) via GPIO pin

3.3 Hardware Design:

- **Microcontroller:** ESP32 / Arduino-compatible board powered via USB-C (5V).
- **Sensor:** LDR connected in voltage divider configuration with a 10kΩ resistor.
- **Output:** High-brightness LED connected through a current limiting resistor (220Ω/330Ω) to the microcontroller pin.

- **Interface:** Breadboard for quick prototyping and easy connections.

The system works on the principle that the resistance of LDR decreases in bright light and increases in darkness. This change produces a varying analog voltage which is converted to digital value by the ADC. The microcontroller processes this value and controls the LED accordingly.

This modular design makes the system simple, reliable, and easy to implement for demonstration

4.Tools/Components

4.1 Hardware Components:

- Microcontroller Board: ESP32 Development Board (with built-in ADC)
- LDR (Light Dependent Resistor) – Light sensor
- LED – Acting as Headlight
- Resistors: $10\text{k}\Omega$ (for LDR voltage divider), $220\Omega/330\Omega$ (for LED current limiting)
- Breadboard – For circuit prototyping
- Jumper Wires – For making connections
- USB-C Cable – For power supply and programming

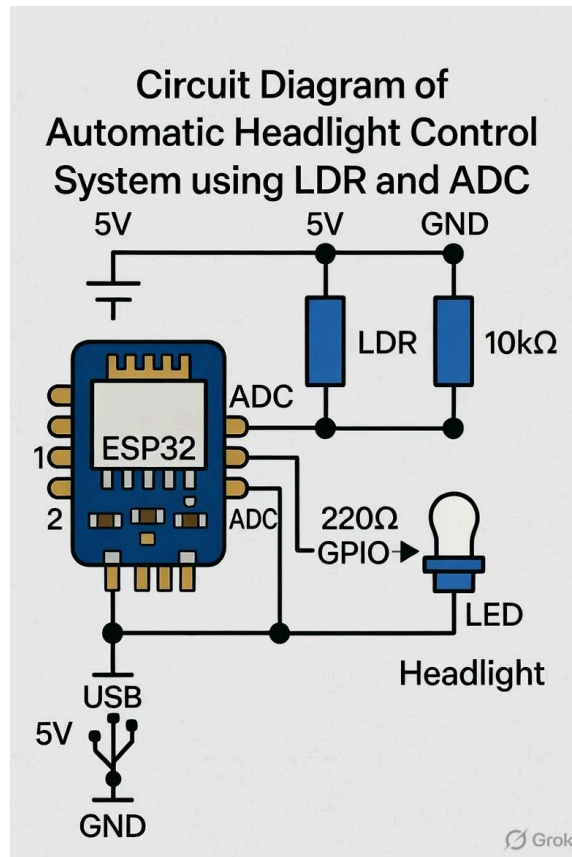
4.2 Software Tools:

- Arduino IDE – For programming the ESP32 microcontroller
- Embedded C Language – For coding

4.3 Other Tools:

- Laptop (for coding and power supply)
- Multimeter (for testing voltage and connections)

5. Circuit operation



The circuit operates on the principle of light intensity detection and automatic switching.

The LDR (Light Dependent Resistor) is connected in a voltage divider configuration with a 10kΩ resistor. It receives 5V supply from the USB-powered ESP32 board. When the surrounding light intensity is high (daylight), the resistance of the LDR is low. This results in a low voltage at the ADC input pin of the microcontroller.

Conversely, when the light intensity decreases (night or dark condition — as shown by covering the LDR with a hand in the demo), the resistance of the LDR increases significantly. This causes a higher voltage drop across the LDR, which is fed to the ADC pin.

The microcontroller continuously reads this analog voltage through its built-in ADC and converts it into a digital value. This digital value is then compared with a predefined threshold value in the program.

- If the ADC value is above the threshold (dark condition), the microcontroller sends a HIGH signal to the output pin, turning ON the LED (Headlight).
- If the ADC value is below the threshold (bright condition), the microcontroller sends a LOW signal, turning OFF the LED.

6. Working

The Automatic Headlight Control System works on the principle of sensing ambient light intensity using an LDR and controlling the headlight accordingly.

The LDR is connected in a voltage divider circuit with a 10kΩ resistor. The resistance of the LDR varies with light intensity:

- In **bright light** (daytime), LDR resistance is low → voltage at ADC pin is low.
- In **darkness** (night time), LDR resistance is high → voltage at ADC pin is high.

This analog voltage is continuously read by the microcontroller through its built-in ADC and converted into a digital value (0 to 4095 in ESP32).

The microcontroller compares this digital value with a predefined threshold:

- If the ADC value **exceeds the threshold** (dark condition), the microcontroller gives a HIGH signal to the GPIO pin, turning **ON** the LED (Headlight).
- If the ADC value is **below the threshold** (bright condition), the microcontroller gives a LOW signal, turning **OFF** the LED
- The **LDR (Light Dependent Resistor)** is the main sensing element. Its resistance changes inversely with light intensity — it has high resistance in darkness and low resistance in bright light.
- The LDR is connected in a **voltage divider circuit** along with a 10kΩ fixed resistor. This configuration converts the resistance change into a proportional analog voltage (0 to 5V).
- This varying analog voltage is fed to the **ADC (Analog-to-Digital Converter)** pin of the ESP32 microcontroller. The ADC converts the analog voltage into a digital value (ranging from 0 to 4095).
- The microcontroller continuously reads these digital values and compares them with a **predefined threshold value** set in the program.

6.1 Code

```
#define LDR_PIN 13 // DO pin from LDR module

#define LED_PIN 23 // LED connected here

void setup() {

  pinMode(LDR_PIN, INPUT);

  pinMode(LED_PIN, OUTPUT);

  digitalWrite(LED_PIN, LOW);

  Serial.begin(115200);

}

void loop() {

  int ldrState = digitalRead(LDR_PIN);
```

```
Serial.print("LDR State: ");  
Serial.println(ldrState);  
if (ldrState == LOW) {  
    // DARK → LED ON  
    digitalWrite(LED_PIN, 1);  
} else {  
    // BRIGHT → LED OFF  
    digitalWrite(LED_PIN, 0);  
}  
delay(300);  
}
```

7. Challenges

During the development of the Automatic Headlight Control System using LDR and ADC, the following challenges were faced:

1. **Sensitivity Adjustment:** The LDR is highly sensitive to light changes. Setting the correct threshold value was difficult because the system was responding even to small variations in room light.
2. **ADC Noise and Stability:** The analog signal from the LDR sometimes contained noise, causing the LED to flicker. This required proper filtering and delay in the code for stable operation.
3. **Lighting Conditions:** Differentiating between actual daylight and artificial room light was challenging during testing.
4. **Breadboard Connections:** Loose connections on the breadboard occasionally caused unreliable behavior during the demonstration.
5. **Limited Power:** Since the prototype is powered only through USB (5V), it cannot drive actual high-power headlights. A relay or transistor driver would be needed for real vehicle implementation.
6. **Calibration:** The system needs recalibration for different environments (indoor vs outdoor, different weather conditions).

8. Output/Results

The Automatic Headlight Control System using LDR and ADC was successfully implemented and tested. The prototype performed as expected, as demonstrated in the project video.

8.1 Key Results:

- The system automatically turns **ON** the headlight LED when the LDR is covered (dark condition).
- The system automatically turns **OFF** the headlight LED when the LDR is exposed to light (bright condition).
- Response time is very fast (less than 1 second).
- The ADC successfully converts the analog signal from the LDR into digital values for accurate decision making by the microcontroller.
- No false triggering was observed during proper threshold setting.
- The entire circuit worked reliably on 5V USB power supply.

8.2 Observation:

When light intensity decreases (LDR covered by hand), the ADC value increases and crosses the set threshold, activating the LED. When light intensity increases, the ADC value drops below the threshold and the LED turns off. This confirms the correct functioning of the automatic headlight control.

The prototype successfully achieves the objective of automatic headlight operation based on ambient light conditions.

9. Applications

The Automatic Headlight Control System using LDR and ADC has several practical and useful applications. Some of the major applications are:

1. **Automobiles / Vehicles:** It can be implemented in cars, bikes, and trucks to automatically control headlights, improving road safety during night, fog, rain, and tunnel conditions.
2. **Street Light Control:** Can be used in smart street lighting systems to automatically turn ON/OFF street lights based on daylight intensity, helping in energy saving.
3. **Home and Office Automation:** Useful in automatic room lighting systems, garden lights, and porch lights.
4. **Industrial Applications:** Can be used in factories and warehouses for automatic lighting control to reduce electricity consumption.
5. **Smart Cities:** Part of intelligent lighting infrastructure to optimize energy usage.
6. **Educational Projects:** Widely used as a beginner-level embedded systems project for students to understand sensors, ADC, and microcontroller interfacing.

10. Conclusion

The Automatic Headlight Control System using LDR and ADC was successfully designed, implemented, and tested. The project demonstrates an efficient and reliable method to automatically control headlights based on ambient light intensity without any manual intervention.

By using an LDR sensor and the built-in ADC of the microcontroller, the system effectively detects darkness and turns ON the headlight LED, and turns it OFF during daylight. The prototype worked as expected, as shown in the demonstration video, where the LED responds instantly to changes in light conditions.

This project helps in enhancing road safety by reducing driver errors and also contributes to energy conservation by preventing unnecessary usage of headlights. It is a simple, low-cost solution that can be further developed for real-time vehicle applications.

Overall, the project provided valuable practical knowledge in sensor interfacing, ADC conversion, microcontroller programming, and embedded system design.

Demo

